

Medical Reasoning with Supervised Fine-Tuning and Group-Relative Policy Optimization

Anubhav De (ade11) Yuanyi Zhang (yzhan795) Xuemeng Wei (xwei40)

December 14, 2025

Abstract

Our group proposed a two-stage training pipeline for improving clinical reasoning in medical reasoning language models. Our system first applies Supervised Fine-Tuning (SFT) by fine-tuning a DistilGPT2 model with 82M parameters on the FreedomIntelligence medical-o1-reasoning-SFT dataset, which contains 500 questions and response pairs to produce explicit, step-by-step medical reasoning. In the second stage, we apply Group-Relative Policy Optimization (GRPO), a reinforcement-learning method that uses a multi-component reward function incorporating semantic consistency, clinical terminology accuracy, reasoning quality, and response structure to further optimize the SFT model.

This combined approach leads to an additional 67% BLEU score improvement compared to SFT alone, while remaining computationally efficient enough to train on a GPU (RTX 4060, 8GB). This combined approach yields measurable improvements in BLEU and ROUGE-L relative to SFT alone and offer a practical way toward deploying improved medical AI systems.

1 Background

Recent advances in large language models have demonstrated strong capabilities for medical reasoning, while reinforcement learning approaches have proven effective for refining model outputs through reward-based optimization. Inspired by work like ReTool, which shows RL can teach LLMs to dynamically select tools for mathematical problem-solving, we hypothesize RL could similarly enable adaptive diagnostic test selection in medicine—balancing accuracy against cost constraints.

The project we earlier proposed was an automated chest X-ray diagnostic impression generated with a multimodal vision-language model enhanced through reinforcement learning. We started with the MIMIC-CXR-JPG dataset, a large-scale, publicly available collection of chest radiographs, together with radiologist-written clinical report, which includes fields including techniques, comparison, findings, and impressions, but varied among all patients. We fine tuned the Qwen2VL model using LoRA (Low-Rank Adaptation) adapters to enable task-specific learning while preserving the general capabilities of the base model. The pipeline began by loading Qwen2VL with LoRA modules and preprocessing chest X-ray images, supporting both DICOM and JPG formats. We normalized pixel values and combined PA and lateral views if both are provided. The reward function we were using included an accuracy reward and a format reward. The true label were gained from CheXpert labeling from the MIMIC-CXR-JPG dataset, which includes 14 diagnostic categories such as No Finding, Cardiomegaly, Pneumonia, Atelectasis, Pleural Effusion, among others. The F-1 score as the accuracy reward was computed by comparing the predicted label extracted from the generated impression from the model and the true label. The formatting rewards

contain text similarity between the generated and reference impressions. We aimed to use this reward function to train the model with GRPO.

The project encountered several critical bottlenecks and failures that severely limited performance and prevented meaningful evaluation. One of the main issue is the image preprocessing itself proved problematic, with many DICOM files either missing critical metadata headers, containing corrupted image bytes, or having path references in the metadata that didn't correspond to actual files on disk, causing silent failures in the image loading pipeline. And we found that GRPO for multimodal LLM requires custom data collection and the current support for image input in GRPO is specific to standard vision-language data. Because of this, we then switch to a medical task that only contains text input.

2 Related Work

Prior work has demonstrated that prompting language models to produce intermediate reasoning steps improves performance on complex tasks. Chain-of-thought prompting [1] showed that explicit step-by-step reasoning enhances accuracy across arithmetic, commonsense, and symbolic reasoning benchmarks. This highlights the possibility of structural guidance in eliciting reliable reasoning, especially in medical problems where interpretability is essential.

DeepSeek-R1 showed that RL-only training can produce strong reasoning behaviors like self-correction [2]. However, their approach requires enormous computational resources, which are impractical in our setting. Our work differs in that we did not abandon the use of an SFT model as a pretraining step for two reasons. First, since we are dealing with medical problems and need accuracy, SFT grounds the model in verified medical knowledge before RL refinement, which could help reduce the risk of harmful hallucinations. Second, we assume that starting from a medically informed checkpoint could let us achieve good reasoning quality with far less RL data and compute.

As mentioned in the DeepSeek-R1 paper [2], policy gradient methods for language model optimization have significantly advanced over the last few years, allowing for more flexible and targeted training. Group Relative Policy Optimization (GRPO) and related methods [3] have shown success in refining model outputs via reward-based learning by normalizing rewards within small groups of generated trajectories, reducing reward estimate variance. In their work, they proposed a combination of reward functions, which guide policy optimization, especially when carefully weighted combinations of different reward components allow for balancing accuracy, comprehensiveness, clarity, and other desirable properties. In medical AI, classical supervised fine-tuning alone can indeed teach the model medical facts like symptoms, but often fails to provide robust reasoning or prevent confident but incorrect conclusions. We therefore combine SFT for foundational medical knowledge with GRPO for quality refinement via a multi-component reward function (semantic similarity, terminology accuracy, reasoning explicitness, and response completeness). Our approach shows that even with modest models and datasets, substantial improvements in medical reasoning are possible under a careful training design.

3 Dataset and Exploratory Analysis

This work will use the FreedomIntelligence medical-o1-reasoning-SFT dataset, a large-scale collection of medical reasoning examples containing 90,120 pairs of (question, response) samples sourced from verified medical knowledge. Each example consists of a medical clinical scenario presented as a question and a corresponding human-constructed step-by-step reasoning chain that culminates in a diagnosis or set of diagnostic possibilities. The dataset was generated using GPT-4o to create

detailed reasoning chains for verified medical problems, ensuring high-quality supervision signals. The dataset was selected specifically because it explicitly contains chain-of-thought style reasoning rather than short answers, providing rich supervision signals for training models to generate detailed medical reasoning. We selected 500 samples from the full dataset for training and evaluation, providing a manageable yet substantial subset for resource-constrained experimentation. This sample size balances two competing concerns: sufficient data to train meaningful medical reasoning capabilities while maintaining tractability on consumer-grade hardware with limited memory. The dataset includes multiple language variants (English, Chinese, and mixed), with our experiments focusing on the English configuration.

4 Model Architecture

Our system is built on a pretrained autoregressive transformer language model (DistilGPT-2), selected to balance reasoning capability and computational efficiency. The model consists of a stack of Transformer decoder blocks with masked self-attention and feed-forward layers, trained using next-token prediction. Given a medical question x , the model generates a reasoning sequence $y = (y_1, \dots, y_T)$ autoregressively:

$$P_\theta(y \mid x) = \prod_{t=1}^T P_\theta(y_t \mid y_{<t}, x) \quad (1)$$

This formulation enables the generation of long, structured reasoning chains conditioned on clinical prompts.

In the first training stage, the pretrained model is fine-tuned using supervised learning on paired medical questions and reference reasoning chains. The input question and target reasoning are concatenated into a single sequence, and aim to minimize the SFT loss:

$$\mathcal{L}_{\text{SFT}} = -\frac{1}{T} \sum_{t=1}^T \log P_\theta(y_t \mid y_{<t}, x) \quad (2)$$

Following SFT, the model is further refined using Group Relative Policy Optimization (GRPO). The SFT-trained model serves as the policy π_θ , while a frozen copy of the same checkpoint is used as a reference policy π_{ref} to stabilize optimization.

For each input question, the policy generates a small group of K candidate reasoning trajectories. A scalar reward is computed for each trajectory, and advantages are calculated relative to the group mean, reducing variance and encouraging comparative improvement.

The GRPO objective is defined as:

$$\mathcal{L}_{\text{GRPO}} = -\mathbb{E}[A_i \log \pi_\theta(y_i \mid x)] + \beta_{\text{KL}} \text{KL}(\pi_\theta \parallel \pi_{\text{ref}}) - \alpha H(\pi_\theta) \quad (3)$$

where A_i denotes the advantage(reward baseline difference), β_{KL} controls deviation from the reference policy, and $H(\pi_\theta)$ is an entropy regularization term encouraging exploration.

The reward function is a multi-component scoring mechanism designed to reflect expert evaluation of medical reasoning. It combines four normalized components. The similarity shows the cosine similarity between the generated and true reasoning. Medical accuracy is computed by medical terminology keywords coverage. Reasoning reward based on the presence of diagnostic and causal reasoning patterns. The length part of the reward encourages responses within a reasonable range.

The final reward is computed as a weighted sum:

$$R_{\text{combined}} = 0.3 R_{\text{similarity}} + 0.3 R_{\text{accuracy}} + 0.2 R_{\text{reason}} + 0.2 R_{\text{length}} \quad (4)$$

5 Training Setup

Policy/backbone. DistilGPT2 (82M parameters; 6 layers, 12 heads, hidden size 768) for fast iteration on consumer hardware.

Stage 1: SFT. We use cross-entropy objective with AdamW as the optimizer with a learning rate 5×10^{-5} (10% warmup), batch size 8, 3 epochs, gradient clip 1.0. Diagnostics in Figure 1. Aggregate SFT results appear in Table 1.

Metric	Initial	Final	Change	Note
Training Loss	3.0984	2.8514	−7.97%	stable descent
Training Perplexity	22.16	17.31	−22%	healthy
Validation Loss	3.4144	2.8738	−15.8%	matches train
Validation Perplexity	30.40	17.61	−42.1%	key metric

Table 1: SFT outcomes on the reasoning subset (90/10 split). Perplexity < 20 indicates good language adaptation.

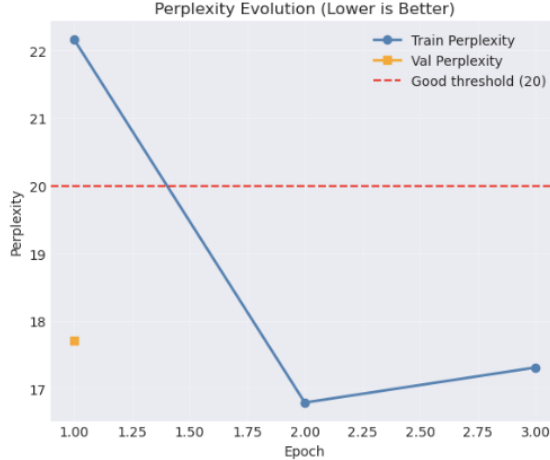


Figure 1: SFT diagnostics. perplexity trends (*lower is better*).

Stage 2: GRPO. We warm-start the GRPO policy from the SFT checkpoint, keep a frozen SFT snapshot as the reference policy for KL control, and optimize the multi-component reward. GRPO was trained with small batches and mixed precision, resulting in stable reward trajectories under our computational constraints. (Figure 2).

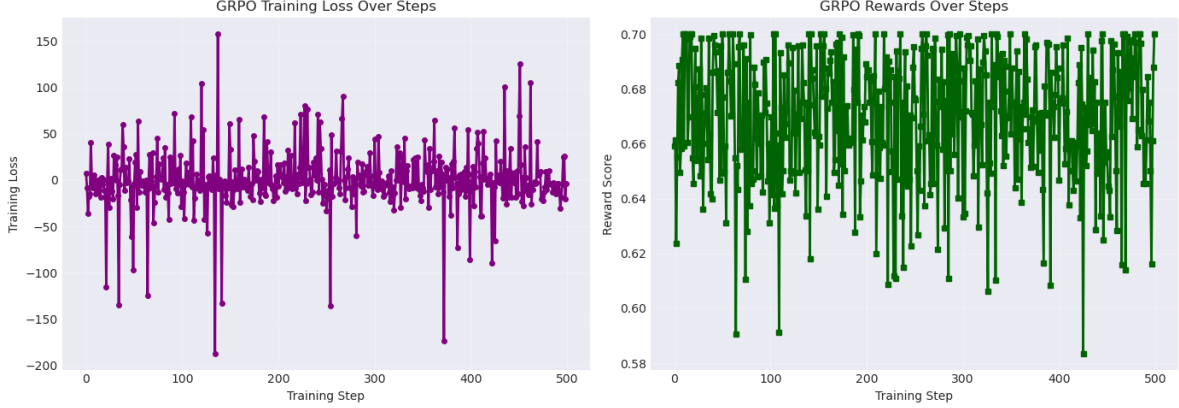


Figure 2: GRPO training dynamics. Loss oscillates as expected for policy optimization; reward stabilizes in a high band (mean ≈ 0.6695 , std ≈ 0.023).

6 Results

6.1 Language Modeling Quality

Figure 3 shows validation loss and perplexity distributions and trends. SFT delivers the large perplexity reduction; GRPO maintains similar values, indicating RL alignment did not harm, but also did not improve next-token accuracy. The original loss of the baseline is 3.4144, and the SFT and GRPO improve to 2.8684.

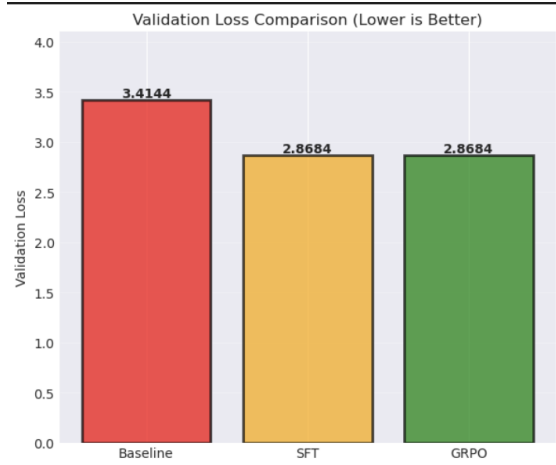


Figure 3: Validation loss/perplexity and distributional views.

6.2 Generation Quality and Efficiency

We choose BLEU and ROUGE-L to compare and contrast the performance of the three models. BLEU (Bilingual Evaluation Understudy) measures precision by counting matching n-grams between generated and reference text, emphasizing exact word matches. ROUGE-L, on the other hand, is based on the longest common subsequence and is useful for capturing structural similarity. The figure below shows that the BLEU score for GRPO is 0.0110, an increase of 0.0054 compared to the SFT-only model. The ROUGE-L score improves by 0.0244 compared to SFT. The average

response length is also longer than that of SFT, which suggests more complete responses, though it deviates from the true response-length distribution, which has a mean of 312.6 tokens.

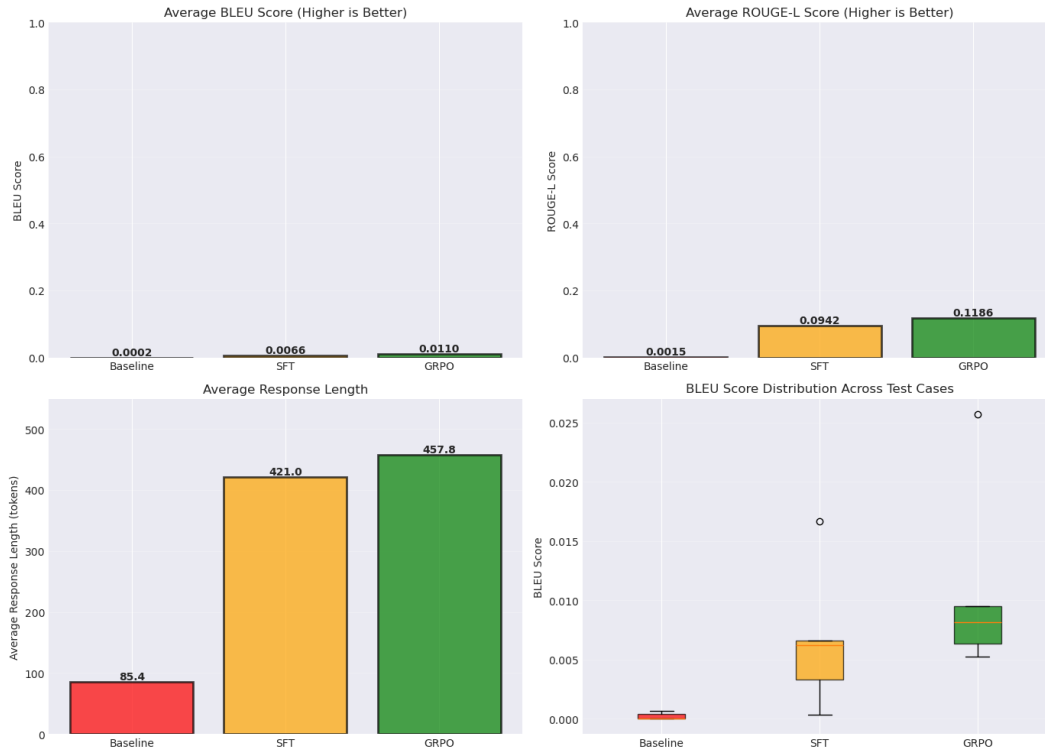


Figure 4: BLEU/ROUGE-L improvements and response length statistics. GRPO increases BLEU and ROUGE-L versus SFT; responses become longer and structured.

Wall-clock. Total ≈ 39.27 minutes (SFT ≈ 1.2 min, GRPO ≈ 38.1 min) on a single RTX 4060 (8 GB).

7 Limitations and Challenges

Several limitations in the dataset affect how well our findings can generalize. First, we trained on only 500 examples out of the full 90,120 samples. This small sample size makes the results highly sensitive to individual examples. If a single difficult case could skew the metrics. Second, all our data come from one source (FreedomIntelligence), meaning the model may simply learn patterns specific to that dataset instead of developing broadly applicable medical reasoning.

We also lack cross-dataset comparison—we did not test on established medical QA benchmarks such as MedQA or PubMedQA—so we cannot say whether the model would perform well outside this particular collection. More importantly, the dataset itself was generated by GPT-4o, which was not reviewed by a doctor. Although the underlying medical problems are well defined and able to check the correctness, the reasoning steps could contain errors, and the model could learn incorrect diagnostic logic.

Our evaluation metrics (BLEU, ROUGE-L) only measure how closely the output matches the reference text at a surface level. A response could score highly on these metrics while being medically wrong or even unsafe. Ultimately, we think either domain-expert review or carefully

designed LLM-based evaluation steps to assess whether the model’s reasoning is valid, accurate, and safe for real-world use.

The limited performance gains observed when transitioning from supervised fine-tuning (SFT) to GRPO can be partially attributed to hyperparameter choices made for computational efficiency. In particular, we used a small group size ($K = 2$), such that only two trajectories were sampled per prompt to compute the group-relative advantage. With such a small group, the baseline reward estimate is highly sensitive to stochastic variation, leading to noisy advantage estimates and reduced stability in policy gradient updates. This increased variance in optimization likely hindered the effectiveness of GRPO, helping to explain why it did not consistently outperform SFT in our experiments.

8 What we learned and Future Work

We learn a two stage SFT, then GRPO could potentially be effective for refining medical reasoning problem. The supervised fine tuning could be used as a cold start, and reinforcement learning with a multi-component reward function could be used to improve the response quality in terms of completeness and structure.

In the future, we will proceed along two complementary directions. First, we will continue improving the current text-only SFT+GRPO medical reasoning pipeline by scaling training to the full 90K-sample dataset, increasing group size to reduce advantage variance, and refining reward design to better balance correctness and verbosity. While automatic metrics (BLEU) provide limited insight into clinical validity, future work will include using a stronger medical LLM as a structured evaluator along predefined clinical criteria. Second, we will revisit the multimodal chest X-ray diagnostic impression generation task that was not fully realized in this project. Planned improvements include more robust DICOM preprocessing, stricter data validation, and more stable multimodal inference using vLLM to support efficient text generation.

9 Conclusion

This work presents a comprehensive two-stage training pipeline that combines Supervised Fine-Tuning with Group-Relative Policy Optimization for medical reasoning generation. This pipeline achieved improvements in all metrics: SFT decreases perplexity by 42.1% ($30.40 \rightarrow 17.61$), while GRPO increases BLEU by 67% ($0.0066 \rightarrow 0.0110$) and ROUGE-L by 26% ($0.0942 \rightarrow 0.1186$). Response length has shown that RL encourages more verbose and structured reasoning outputs. Critically, the full pipeline was trained for 39.27 minutes on consumer hardware (RTX 4060, 8GB VRAM), demonstrating the practical feasibility of advanced training techniques without enterprise infrastructure.

References

- [1] Wei, J., Wang, X., Schuurmans, D., et al. (2022). Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems* (NeurIPS).
- [2] DeepSeek-AI, Guo, D., Yang, D., et al. (2025). DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. *arXiv preprint arXiv:2501.12948*.
- [3] Christiano, P. F., Leike, J., Brown, T., et al. (2023). Deep reinforcement learning from human preferences. In *International Conference on Machine Learning (ICML)*.

- [4] Johnson, A. E. W., Pollard, T. J., Berkowitz, S. J., et al. (2019). MIMIC-CXR-JPG: A large publicly available database of labeled chest radiographs. PhysioNet. Available at: <https://physionet.org/content/mimic-cxr-jpg/2.1.0/>.

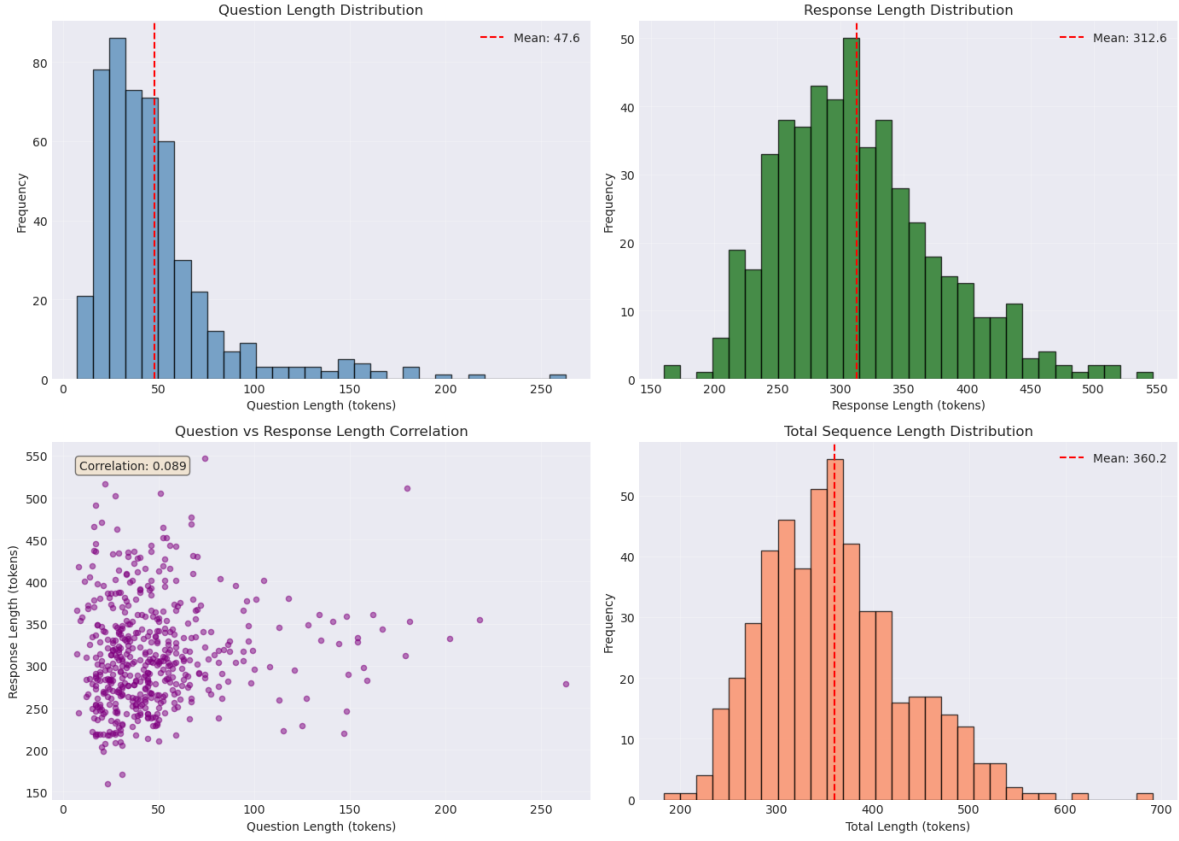


Figure 5: Dataset characteristics. Mean question length 47.6 tokens (95th percentile 115.1), mean response length 312.6 (95th percentile 431.1), mean total 360.2 tokens. Question–response length correlation is weak.